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Digital Twin-Driven Optimization of Water Production in Gathering Centers: A Koc North Kuwait Case Study

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Abstract

Increasing water production in a mature asset is one of the critical issues facing most fields around the world. Subsurface uncertainties and poor water injection strategies have led to a significant and unintended increase in water-cut. This paper will discuss in detail how a digital twin of the well and surface network model was utilized to significantly reduce water production while maintaining or increasing oil production. Custom-built workflows were created, employing advanced algorithms to identify optimum criteria, such as changing parameters in artificial lift, wellhead chokes, or closing low oil-producing wells. The algorithm performed several thousand iterations, taking into consideration water handling costs and current oil prices while optimizing production.

A dedicated Automated Water Production Optimization (AWPO) workflow was created to achieve the goal of minimizing water production. This workflow is part of a larger system that integrates well modeling, surface network modeling, calibration of chokes and surface pipeline networks. Custom-built workflows check the quality of well models and real-time data. The input to the AWPO workflow is a calibrated surface network model. A simulated annealing algorithm is utilized to optimize water production. The algorithm runs several thousand iterations to adjust artificial lift parameters, choke settings, and even closing of wells while maintaining current oil production rates or slightly reducing oil production, but only when the ratio of water handling cost to current crude oil price is significantly high. This means that even though oil production may slightly reduce, there will be a large reduction in water handling costs. The workflow ensures that all operational constraints, such as pumps being within the operational envelope, motor load within set tolerable limits, and flowing bottomhole pressure above bubble point pressure, are honored. Current, optimized, and total costs of optimization are calculated at the back end.

The AWPO workflow runs on a daily basis for several gathering centers, accounting for several thousands of barrels of oil production and over a million barrels of water production per day. Current runs have shown that AWPO recommendations result in a reduction of several thousand barrels of water production and save thousands of dollars in water handling costs. The workflow plays a significant role by providing specific actions to the field development and production operations teams on which wells to target and what actions to take to reduce water production.

Many existing methodologies have mainly concentrated on targeting the subsurface domain or completion technology to reduce water production. The approach used in this paper, which involves utilizing custom-built automated workflows and algorithms with a digital twin of the well and surface network model to minimize water production, is a pioneering idea.

Problem Statement

Major oil fields around the world are increasingly dealing with assets which are producing high rates of water. As the fields mature, the complexities of managing water production become more pronounced. One of the primary reasons for this issue is subsurface uncertainties, which can make it difficult to predict and control the movement of water within the reservoir. These uncertainties often lead to inefficient water management practices, exacerbating the problem.

Addressing this problem requires innovative approaches that go beyond traditional methods. This paper explores how a digital twin of the well and surface network model was employed to significantly reduce water production while maintaining or even increasing oil production.

The digital twin approach involves creating a virtual replica of the physical assets, allowing for real-time simulation and optimization. Custom-built workflows were developed, incorporating advanced algorithms to identify the optimal criteria for reducing water production. These criteria included adjusting parameters in artificial lift systems, modifying wellhead choke settings, and selectively shutting down low oil-producing wells. By simulating various scenarios, the digital twin provided valuable insights into the most effective strategies for minimizing water production.

Methodology

The Automatic Water Production Optimization (AWPO) workflow is an integral component of the Integrated Production Optimization (IPO) workflow (Figure 1), which serves as an end-to-end surface network management ecosystem. The IPO workflow begins by converting Prosper Well Models, validated with the latest well test data, into a format compatible with the Surface Network Simulator, specifically the Nexus Simulator. This task is handled by the Automatic IPR VLP Generation and Conversion (AIVGAC) workflow. The AIVGAC workflow generates, converts, and saves the Inflow Performance Relationship (IPR) and Vertical Lift Performance (VLP) data in text files with keywords that the Nexus Simulator can read.

Every day at 5 AM, the Automatic Network Model Builder (ANMB) workflow is scheduled to run. This workflow collects all necessary data to create or update the surface network model to replicate the current field conditions. The data required for this process is sourced from various corporate databases of the Kuwait Oil Company (KOC). The ANMB workflow updates well names, well status, surface network configurations (such as well to manifold inlet slot, well to gathering center slot, and gathering center slot to header connections), real-time data (including wellhead pressure, flowline pressure, artificial lift parameters, choke settings, manifold pressure, and gathering center header pressure), and well test data. It is important to note that the initial surface network model is built manually, and the ANMB workflow uses this as a base, updating it based on the current data. The workflow incorporates various logics to account for data issues, ensuring the accuracy and reliability of the model.

The data from the KOC databases is converted into a standard format before being used by the ANMB workflow through Extract, Transform, and Load (ETL) programs. The output of the ANMB workflow is an updated model in a format consistent with the Nexus Simulator. This transformation is carried out by the Digital Field Solver Workflow, which uses custom-built Python codes to perform the conversion. The IPR and VLP curves generated by the AIVGAC workflow are integrated into this process as well.

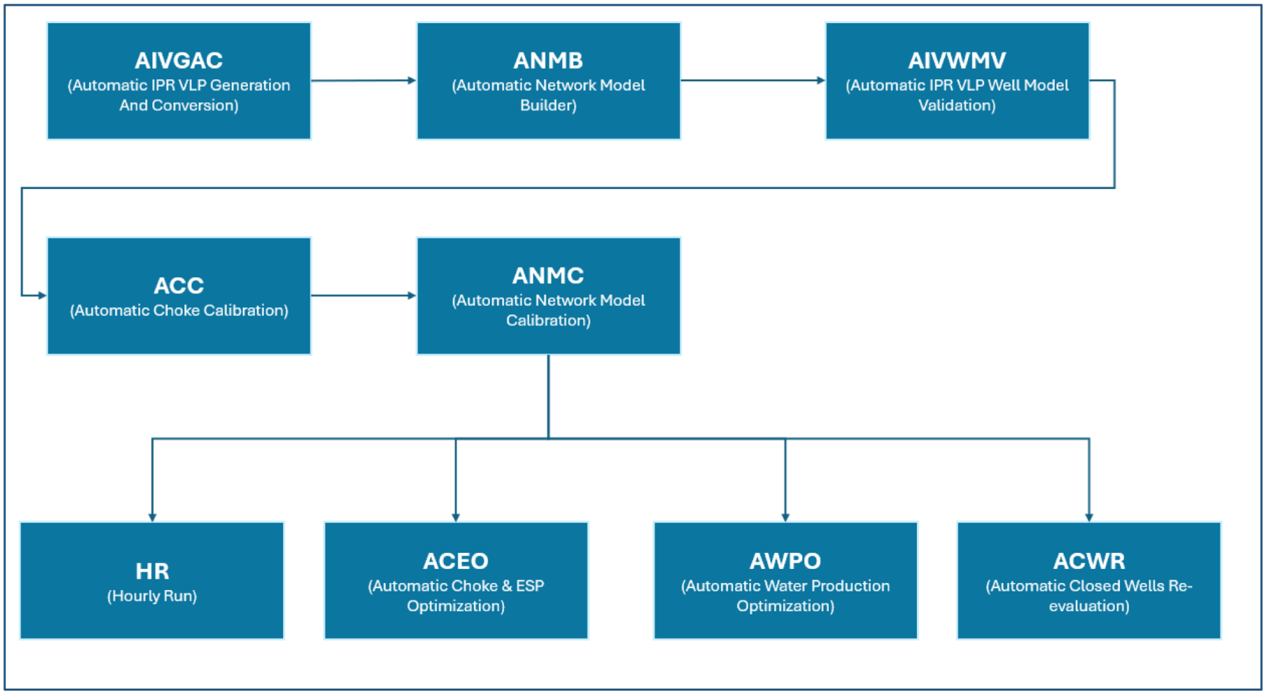

Figure 1—*Integrated Production Optimization (IPO) workflow*

The Automatic IPR VLP Well Model Validation (AIVWMV) workflow tests the quality of the well model calibration. The surface network model created by the ANMB workflow is deactivated after the wellhead connection. Individual wells are subjected to their well test conditions with artificial lift parameter and wellhead pressure constraints, and a simulation is carried out. The reported liquid rates are compared with the well test results. If the deviation is greater than 10%, the well models are sent back for re-calibration. These wells are not considered for AWPO optimization.

The next step is the Automatic Choke Calibration (ACC), where the model generated at the end of the ANMB workflow is again deactivated after the choke connections. Wells that have passed the AIVWMV process are subjected to well test conditions with artificial lift parameter, choke setting, and flowline pressure constraints. If the well can reproduce the wellhead pressure observed during the test conditions, the choke is considered calibrated. Otherwise, a calibration process is carried out to identify the discharge coefficient that can reproduce the wellhead pressure. If the wellhead pressure is not reproduced within the 10% range, the choke is considered uncalibrated, and these wells are not used in the AWPO workflow for optimization.

The final step before the AWPO workflow is the Automatic Network Model Calibration (ANMC). This process is carried out for wells that have passed the ACC workflow. The ANMC workflow utilizes a simulated annealing algorithm to tune the friction factor coefficients so that the model-calculated pressure data matches the real-time data from the field. At the end of the workflow, the calibrated pipelines are used in the AWPO workflow

Automatic Water Production Optimization (AWPO)

A dedicated Automated Water Production Optimization (AWPO) workflow was developed to minimize water production in oil fields. As described before, this workflow is an integral part of a comprehensive system that includes well modeling, surface network modeling, and the calibration of chokes and surface pipeline networks. The AWPO workflow leverages custom-built processes to ensure the quality of well models and real-time data, providing a robust foundation for optimization efforts.

The AWPO workflow begins with a calibrated surface network model as its input. This model is essential for accurately representing the current state of the field and serves as the basis for optimization. The workflow employs a simulated annealing algorithm, a powerful optimization technique that iteratively adjusts various parameters to find the optimal solution. The algorithm runs several thousand iterations, fine-tuning artificial lift parameters, choke settings, and even considering the closure of certain wells. The primary goal is to reduce water production while maintaining current oil production rates or allowing for a slight reduction in oil output, but only when the ratio of water handling costs to the current crude oil price is significantly high. This approach ensures that any reduction in oil production is offset by substantial savings in water handling costs.

One of the key strengths of the AWPO workflow is its ability to honor all operational constraints. For instance, it ensures that pumps operate within their designated envelopes, motor loads remain within set tolerable limits, and flowing bottomhole pressure stays above the bubble point pressure. These constraints are critical for maintaining the integrity and efficiency of the production system. By adhering to these operational limits, the workflow ensures that the optimization process does not compromise the safety or performance of the equipment.

The AWPO workflow also includes a comprehensive cost analysis component. It calculates the current, optimized, and total costs of optimization at the back end, providing a clear picture of the economic impact of the recommended actions. This cost analysis is crucial for decision-making, as it allows operators to weigh the benefits of reduced water production against any potential decrease in oil output. By providing detailed cost information, the workflow helps operators make informed decisions that maximize overall profitability.

Results & Discussion

The Automated Water Production Optimization (AWPO) workflow operates daily across several gathering centers in North Kuwait, which collectively produce several thousand barrels of oil and more than a million barrels of water each day. This workflow has demonstrated significant effectiveness, with current runs showing that AWPO recommendations can reduce water production by several thousand barrels daily. This reduction translates into substantial cost savings, as it lowers the expenses associated with water handling by thousands of dollars (Figure 2)

Figure 2—*Automatic Water Production Optimization (AWPO) dashboard*

The AWPO workflow plays a crucial role by providing specific recommendations to the production operations teams. These recommendations include optimal settings for Electric Submersible Pump (ESP) frequency and choke adjustments, guiding the teams on which wells to target and what actions to take to minimize water production. By fine-tuning these operational parameters, the workflow helps in achieving a more efficient and balanced production process, reducing unnecessary water extraction while maintaining or enhancing oil output.

Conversely, the AWPO workflow increases the choke settings and ESP frequency for wells with higher potential for oil production. These wells are more efficient in producing oil with less water, making them more profitable. By optimizing the operational parameters for these wells, the workflow enhances their oil production capabilities, contributing to the overall profitability of the gathering center. This strategic adjustment ensures that the production system focuses on maximizing oil output while minimizing water production, leading to a more efficient and cost-effective operation.

The simulation runs thousands of iterations, each time adjusting the parameters to find the optimal settings that maximize the objective function, which in this case is the profitability of the gathering center. The optimization algorithm employed by the AWPO workflow is designed to explore a wide range of possible configurations, evaluating the impact of each adjustment on the overall system performance. This iterative process allows the algorithm to identify the most effective combination of choke settings and ESP frequencies that achieve the desired balance between oil production and water handling costs.

The optimization algorithm is set to run for a maximum of 3600 seconds (one hour). This time constraint ensures that the optimization process is both thorough and efficient, providing actionable recommendations within a reasonable timeframe. By running the algorithm for this duration, the workflow can explore a vast number of potential configurations, increasing the likelihood of finding the optimal solution. The results of the optimization are then used to update the operational parameters for the wells, ensuring that the production system operates at peak efficiency.

The AWPO workflow's ability to dynamically adjust choke settings and ESP frequencies based on real-time data and simulation results is a key factor in its effectiveness. By reducing the operational parameters for wells with high water cut and increasing them for wells with higher oil production potential, the workflow optimizes the overall performance of the gathering center. The use of a simulated annealing algorithm, running thousands of iterations within a set timeframe, ensures that the optimization process is both comprehensive and efficient. This approach not only enhances oil production but also significantly reduces water handling costs, leading to improved profitability and operational efficiency. The AWPO workflow represents a sophisticated and innovative solution for managing water production in mature assets, leveraging advanced algorithms and real-time data to achieve optimal results.

Conclusion

In conclusion, the development and implementation of the Automated Water Production Optimization (AWPO) workflow represents a significant advancement in the management of water production in oil fields. Many existing methodologies for reducing water production have primarily focused on the subsurface domain or completion technology. These traditional approaches often involve modifying the reservoir characteristics or enhancing the completion techniques to control water influx. While these methods have been effective to some extent, they do not fully address the complexities of water production management across the entire production system.

The approach presented in this paper introduces a novel and pioneering idea by leveraging custom-built automated workflows and algorithms in conjunction with a digital twin of the well and surface network model. This integrated strategy goes beyond the subsurface domain, encompassing the entire production network to minimize water production. The digital twin daily updates and calibrates with the latest data from the field. This dynamic and adaptive model allows provides optimum choke settings and ESP frequencies, to optimize Oil production and reduce water production. One of the key strengths of the AWPO workflow is its ability to honor all operational constraints. It ensures that pumps operate within their designated envelopes, motor loads remain within set tolerable limits, and flowing bottomhole pressure stays above the bubble point pressure. These constraints are critical for maintaining the integrity and efficiency of the production system. By adhering to these operational limits, the workflow ensures that the optimization process does not compromise the safety or performance of the equipment. This ability enables operators to make informed decisions and implement strategies that effectively minimize water production.

In summary, the AWPO workflow represents a sophisticated and innovative solution for managing water production in mature assets. By integrating advanced algorithms, real-time data, and comprehensive cost analysis, the workflow delivers significant benefits in terms of both operational efficiency and economic performance. This pioneering approach to water production optimization not only addresses the challenges of subsurface uncertainties and poor water injection strategies but also ensures that oil recovery is maximized, thereby enhancing the overall performance and profitability of oil fields.

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Nomenclature

IPO - Integrated Production Optimization
AIVGAC - Automatic IPR VLP Generation and Conversion
AIVWMV - Automatic IPR VLP Well Model Validation
ACC - Automatic Choke Calibration
ANMC - Automatic Network Model Calibration
ACEO - Automatic Choke and ESP Optimization
AWPO - Automatic Water Production Optimization
ETL - Extract Transform Load
CD - Discharge Co-efficient
UI - User Interface

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